
Student Internship Programme (SIP)

Project Report

at Ebeecare

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by

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Project Title: Web application development and iOS application development

Project ID: S9669050

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Summary

In this project, I have added more features and fixed some UI/UX bugs for the current Ebeecare web application development. The new features includes improvement of data tables for admin accounts to view and edit all the cases at one glances and window local storage to store user's service preferences in the browser and suggest that specific service for the user's subsequent visit to our website. I bind the user's device to the server and set up firebase push notification for Ebeecare's iOS partner application. I have also update and fixed the outdated dependencies in both Ebeecare and EbeecarePartners iOS application. The design criteria and the implementation details are presented in this report.

Subject Descriptors:

- 2. Update dependencies in both Ebeecare and EbeecarePartners iOS applications
- 3. Set up firebase push notification for Ebeecare's iOS partner application
- 4. UI/UX improvements and bugs fixed

Keywords:

- UI/UX React and Redux
- iOS Swift Development

Implementation Software and Hardware:

- Software:
 - Xcode
 - Sublime Text
 - Terminal
 - SourceTree
 - BitBucket (Code repository)
 - JIRA (Track SCRUM issues progress)
- Hardware:
 - iPhone
 - iPad
 - Macbook (for web browser)

Acknowledgement

I would like to acknowledge the company's CTO, Eric Lu, as he has help me through these 12 weeks, to set up the working coding environment and to understand the legacy code written by previous developers.

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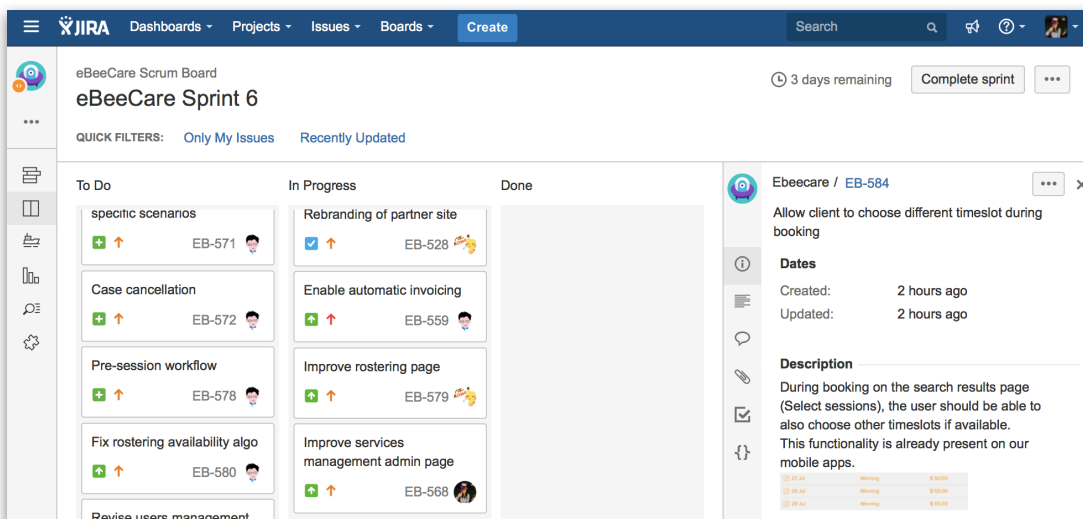
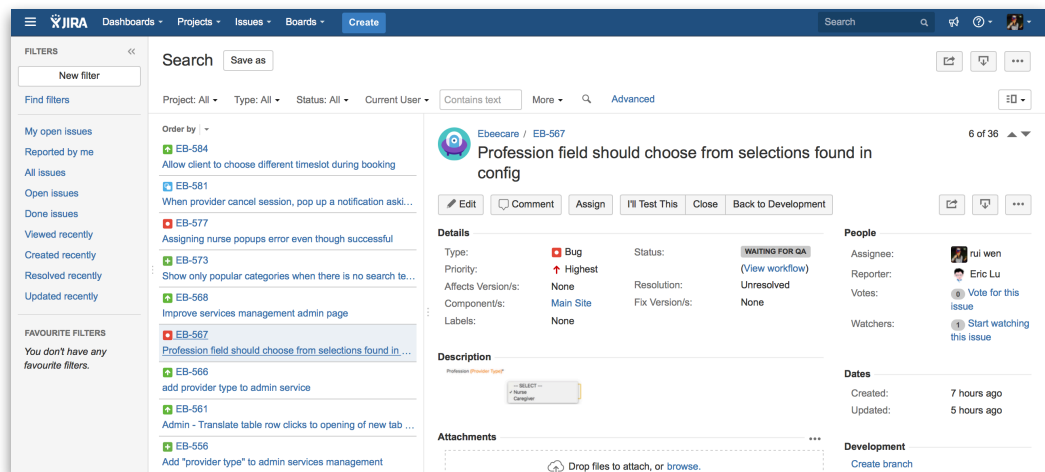
Main Report

1. Introduction

Ebeecare is an online/mobile platform to help caregivers to search, book, manage and pay for home healthcare services in an accessible and flexible way. Our innovative and responsive algorithm will match their needs, location and timing with the most suitable healthcare professional available at the lowest cost. Unlike traditional nursing agencies, Ebeecare provides patients a direct access to nurses, caregivers, therapists and doctors, without having to navigate through the complicated and fragmented healthcare market place.

Ebeecare uses **Single Page Applications (SPAs)** framework for its web application platform. On a SPA, after the initial page load, no more HTML gets sent over the network. Instead, only data gets requested from the server (or sent to the server). So while a SPA is running, only the part that is being changed gets sent over the wire, which takes a lot less time and bandwidth than constantly sending HTML. React is one of the major sites using React in production are Facebook (including Instagram), Airbnb, and Uber.

Ebeecare uses **JIRA** to track issues and weekly SCRUM sprints. The IT department also have daily SRCUM meeting at 11am to update the SCRUM master on our sprint progress.



JIRA is also used to create and merge branch which is in sync with BitBucket's code repository.

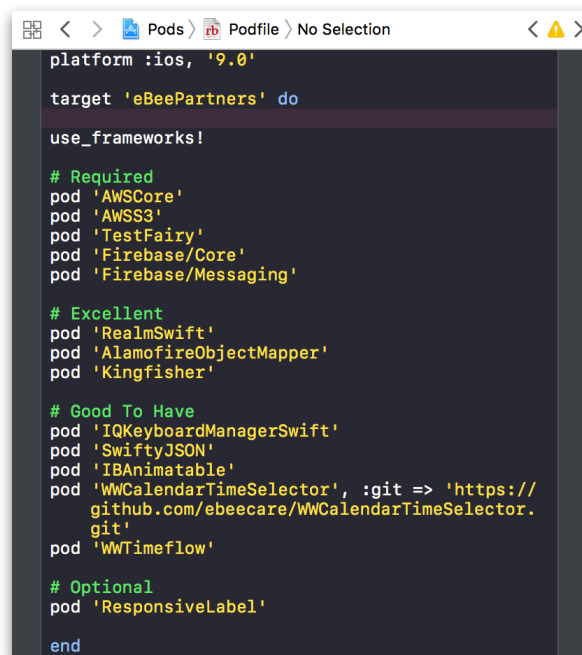
2. Update dependencies in both Ebeecare and EbeecarePartners iOS application

Issues/Problems:

The WWCalendarTimeSelector dependency used by the previous developer as shown in the right image, is obsolete and it is causing runtime errors. The programmer for this specific dependency is no longer updating the dependency.

Solution:

Fork the original code from the respective git repository into Ebeecare's github account, fixed the faulty part of the code and ensure both Ebeecare and EbeecarePartners iOS application uses the new dependency.



```
platform :ios, '9.0'

target 'eBeePartners' do

  use_frameworks!

  # Required
  pod 'AWSCore'
  pod 'AWSS3'
  pod 'TestFairy'
  pod 'Firebase/Core'
  pod 'Firebase/Messaging'

  # Excellent
  pod 'RealmSwift'
  pod 'AlamofireObjectMapper'
  pod 'Kingfisher'

  # Good To Have
  pod 'IQKeyboardManagerSwift'
  pod 'SwiftyJSON'
  pod 'IBAnimatable'
  pod 'WWCalendarTimeSelector', :git => 'https://github.com/ebeecare/WWCalendarTimeSelector.git'
  pod 'WWTImeflow'

  # Optional
  pod 'ResponsiveLabel'

end
```

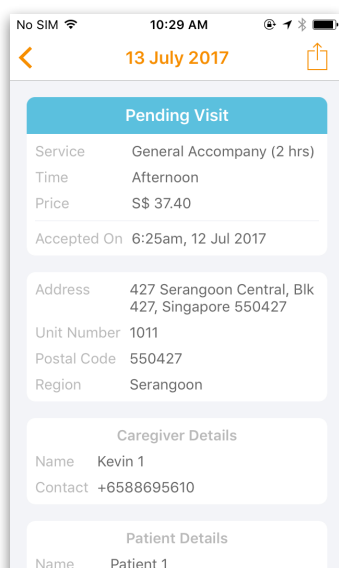
3. Set up firebase push notification for Ebeecare's iOS Partner application

Issues/Problems:

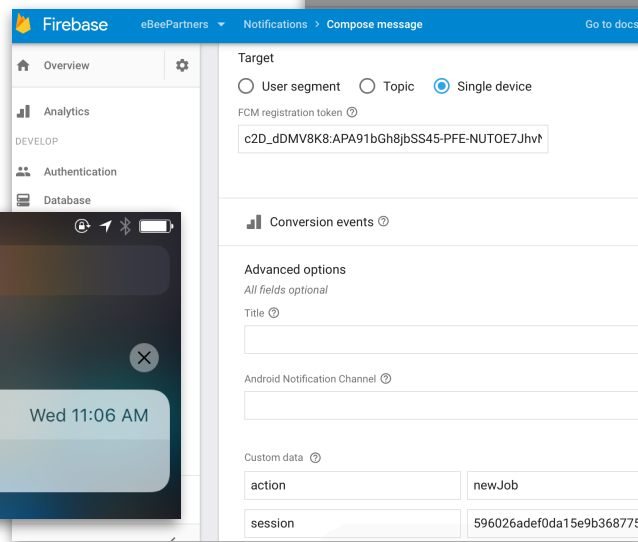
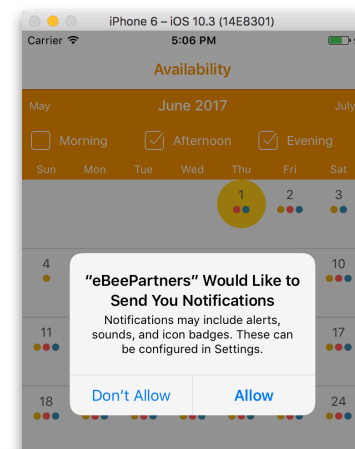
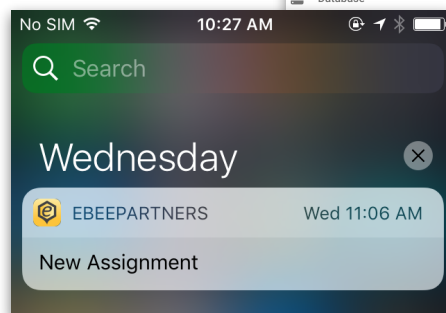
There is a need to mass notify provider (nurse) on new job available whenever a patient book an appointment or whenever an admin gives a job assignment to a nurse.

Solution:

Firstly, we need to get user's device token and Firebase Cloud Messaging (FCM) token and post them into the backend server to verify the authentication of the user. Clicking onto the notification will redirect user to the respective job/assignment session to accept job or view assignment.



the respective job/assignment session to accept job or view assignment.



4. UI/UX improvements and bugs fixed

4.1 Minor UI Improvements

In the first few weeks of internship, I improve the UI of the web app and fixed some basic bugs as I start to understand and modify the load of legacy codes.

These minor issues/improvements include:

- Image uploaded by client cannot be clicked on to view when Nurse accepts the case. Only thumbnail is available.
- Improve CSS for better view (bold the headers and prevent UI component column to overflow and spill over to the next column).
- Data validation for patient and care provider sign up and addition forms.
- Add Chinese subtitles to company videos and include them in the 'About Us' page.
- iOS 'Account Activation' alert popup is not removed despite the user being activated already.

4.2 Patient's Condition Documentation

The care provider are required to log the patient's condition documentation after every visit before they get their payout.

The image displays two screenshots of a web application for 'Provider Documentation'.

Left Screenshot (Step 1 of 4): This screen shows the 'Overall Assessment' form. It includes a navigation bar with steps 1 (Patient Assessment), 2 (Procedural Assessment), 3 (Summary of Findings), and 4 (Confirmation). The 'Overall' tab is active. Below the navigation bar, there's a section for 'Overall' with a description and a note: 'It is compulsory that you complete this Overall Assessment'. The form is divided into three columns: 'Any Problems', 'Evidence', and 'Interventions/Evaluation'. Each column has a table with rows for different conditions (e.g., Sleep Problem, Problems with eating and feeding, Incontinence) and a 'Yes/No' switch.

Right Screenshot (Step 2 of 4): This screen shows the 'Add More' section. It lists three forms to be added: 'Bate Jensen Wound Form (Bate)', 'Nasogastric Tube (NGT)', and 'Urinary Catheter'. Each form has a description and a '+Add Form' button. At the bottom, there's a 'Form Added' button.

4.2.1 Fix Faulty Yes/No Switch In Most Documentation Forms

4.2.2 Add Delete Button For Optional Forms

Previously, all added forms must be fully filled before a nurse can submit the documentation, however there are cases where the nurse accidentally add unnecessary irrelevant forms and hence could not fill it up and submit the documentation. Hence I added the delete button for optional forms.

Medications

	Route	Medication	Dose	Unit	Cycle	Duration	Unit	Instructions
1.	(IV) ▾	Paracetamol	2	unit ▾	6H, ▾	1.5	Day ▾	Consume after meal
2.	--- ▾			--- ▾	--- ▾		--- ▾	(e.g. timing)
3.	--- ▾			--- ▾	--- ▾		--- ▾	(e.g. timing)

[+ Add Row](#)

Vital Signs

Blood Pressure / (mmHG)

Low: < 90/60
Normal: < 120/80
High: > 140/90

☐ Unable to assess

Blood Glucose Level (mmol/L) ↔ (mg/dl)

High: > 10 mmol/L

☐ Unable to assess

Temperature (°C)

High: > 37.5°C

☒ Unable to assess

Heart Rate per minute

Infant: 100 to 160 bpm
Children: 70 to 120 bpm
Adult: 60 to 100 bpm

☒ Unable to assess

4.2.3 Medication Table In Documentation Was Not Working

4.2.4 Data Validation For Many Fields In The Documentation Forms

4.3 Add Referral Program Into EbeecarePartners iOS Application

Issues/Problems:

My company wanted to increase the number of nurses and patients by implementing a referral program where nurse gets a cash incentive on their next payout when the nurse friend they refer did an assignment with our company.

Solution:

Add a 'Refer A Friend' button into EbeecarePartner iOS application. Clicking on the button will redirect user to Ebeecare Referral Portal.

eBeePartners 10:50 AM refer.ebeecare.com



Ebeecare Referral Portal

Please fill out the form
To go to main page click on <http://www.ebeecare.com>
To register as service provider click on <https://www.ebeepartners.com>

* Required

I want to refer a : *

☐ Patient /Elderly

☐ Partner (Nurse, Doctor...)

No SIM 10:49 AM



Email

Password

[LOGIN](#)

[REFER A FRIEND](#)

[Forget Password?](#) [Sign Up](#)

4.4 Time Slot With Accepted Cases Should Be Cross-Out In The Schedule Calendar

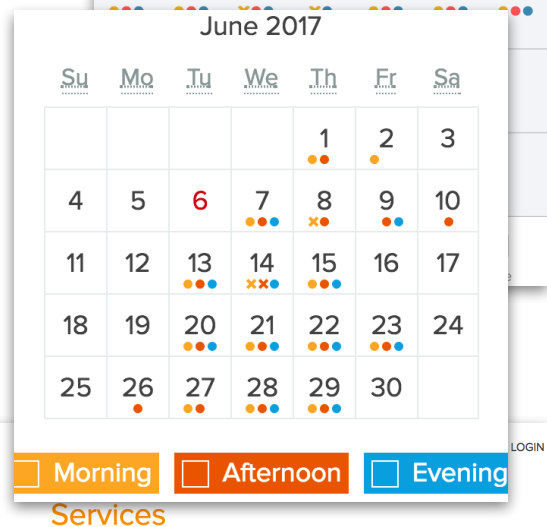
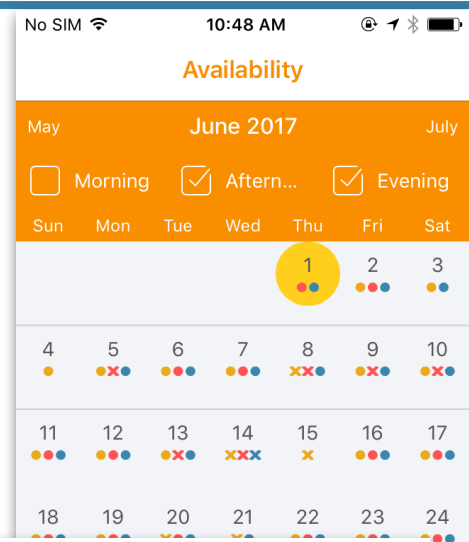
Issues/Problems:

The nurse have to indicate their available time slots in the schedule calendar to view the jobs available in these time slots. However, previously, the time slots with accepted job cases was not displayed differently (all available time slots was indicated with a dot) and they could remove the availability, causing much trouble to the company.

Solution:

The time slots with accepted job cases are indicated with a cross and the user interaction of the checkbox selection button for these cross-out time slots are disabled to prevent nurses from removing their availability for a time slot with accepted cases.

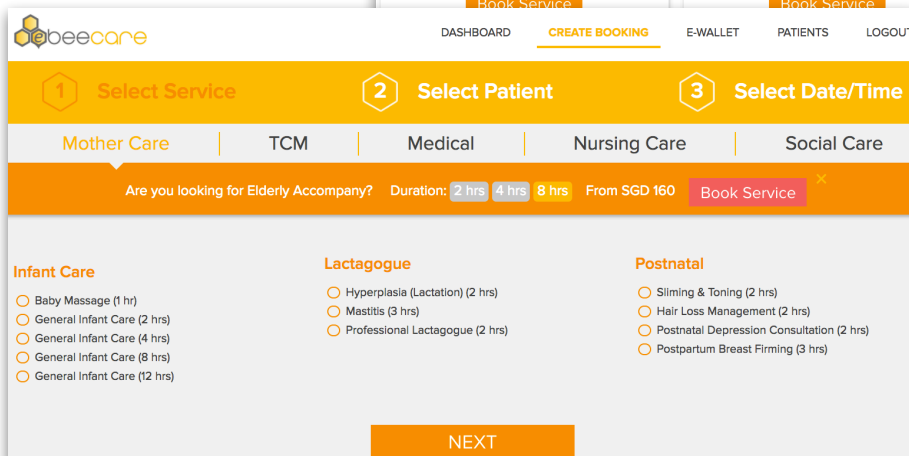
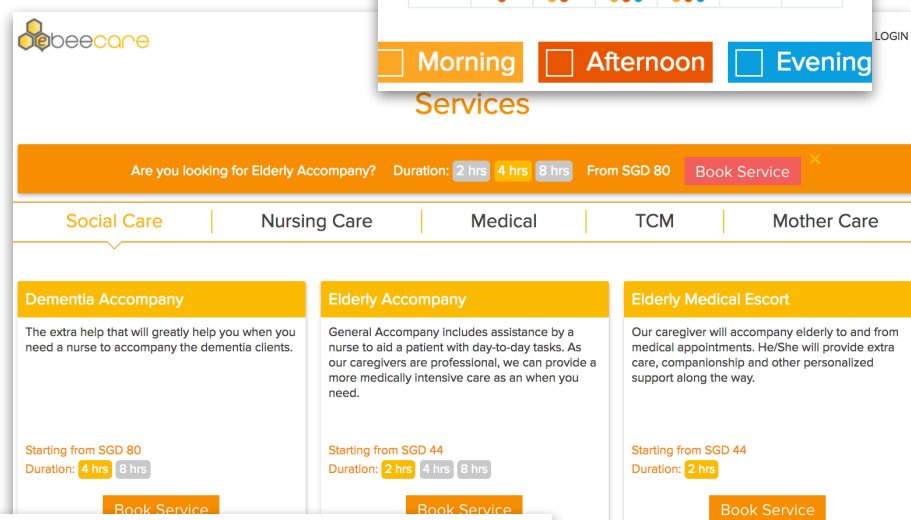
This is done in both the web application and iOS partner's application.



4.5 Provide Service Suggestion According To User Preference

Issues/Problems:

There are too many services available for client to choose from.



Solution:

Use local storage to provide service suggestion according to user preferences from their past service selection. This is done for both ad hoc and logged in user booking.

4.6 Improvement For Admin Management View

Made improvement to most of the tabs in the Ebeecare web application's admin account. For all the tabs that contains tables, I make each row clickable and clicking onto the row will open a new tab which display the details of the selected application/booking/case/transaction/services/category.

I also added filtering and sorting features for some of the tables and added some essential column to boost the productivity of admin user.

5. Conclusion

In conclusion, I have update dependencies in both Ebeecare and EbeecarePartners iOS applications, set up firebase push notification for Ebeecare's iOS partner application and did many UI/UX improvements and bugs fixed throughout these 12 weeks of internship at Ebeecare.

Before I started this internship, I did not have any experience on react, redux and swift. I thought web development consist of HTML, CSS and Javascript only. Little did I know that there would be so many useful frameworks to work with. Throughout the internship, I have gain a lot more experience on react, redux frond-end web development and swift iOS development. I also had the opportunity to do many independent learning from the web, mainly from StackOverFlow, Google and articles from Medium.

Web and iOS development skills are high in demand as every company needs them, however, as a career option, I am more incline towards AI sector instead, thus web and iOS development would be my backup plan.

The screenshot displays the Ebeecare Admin Management interface. At the top, there's a navigation bar with tabs: DASHBOARD, PROMOCODES, APPLICATIONS, ROSTERING, BOOKINGS, CASES (selected), and USERS. Below this is a sub-navigation bar with: TRANSACTIONS, SERVICES, CATEGORIES, SETTINGS, PATCHES, and LOGOUT. The main header shows 'Case Management' and a user profile 'Welcome, Admin'. A filter section includes an 'Alias' dropdown, a 'Filter keyword' input, and a 'clear sort & filter' link. Two checkboxes are present: 'Show only expired sessions which are "pending-visit"' and 'Show only completed sessions where the service provider have not been paid'. A table lists cases with columns: ALIAS, CLIENT, PATIENT, PROVIDER, PRICE, SERVICE, DATE, TIMESLOT, and STATUS. The table contains several rows, with the first row (000623) selected. Below the table, the 'Case Detail' view for Case #000623 is shown. It includes an 'Edit' button and three main sections: CASE DETAILS, CASE LOCATION, and CASE STATUS. CASE DETAILS includes fields for Type, Alias, Service, Original Price, Price, Provider Price, Date, Time, and Note. CASE LOCATION includes fields for Postal, Region, Neighborhood, Address, and Unit. CASE STATUS includes fields for Status, Applications, Paid, Refund, and buttons for Feedback, Refund, Payout, and Cancel. Below these are sections for PATIENT, SERVICE PROVIDER (marked 'Provider is not assigned'), and CLIENT, each with their respective details.

ALIAS	CLIENT	PATIENT	PROVIDER	PRICE	SERVICE	DATE	TIMESLOT	STATUS
000623	Kevin 1	Kevin		\$ 120.00	Mastitis	2017-07-28 (Fri)	morning	awaiting-caregiver
000622	Kevin 1	Kevin	Kevin	\$ 90.00	Professional Lactagogi	2017-07-27 (Thu)	morning	pending-payment
000621	kevin	patient 4		\$ 160.00	Elderly Accompany	2017-07-21 (Fri)	morning	expired
000620	Kevin 1	Eric		\$ 80.00	General Accompany	2017-07-24 (Mon)	evening	expired
000619	Kevin 1	Kevin	Kevin	\$ 176.00	General Accompany	2017-07-29 (Sat)	afternoon	pending-payment
000618								

Case #000623

[Edit](#)

CASE DETAILS

- Type: Registered User [View Booking](#)
- Alias: 000623
- Service: Mastitis (3 hrs)
- Original Price: SGD 120.00 [Edit](#)
- Price: SGD 120.00
- Provider Price: SGD 102.00 [Edit](#)
- Date: 2017-07-28 (Fri)
- Time: Morning
- Note: -

CASE LOCATION

- Postal: 049944
- Region: Outram
- Neighborhood: Outram
- Address: 1 Amoy St, Singapore 049944
- Unit: 100

CASE STATUS

- Status: Awaiting Caregiver [set visited](#)
- Applications: 0
- Paid: No
- Refund: No
- [Go](#) [Feedback](#) [Refund](#) [Payout](#) [Cancel](#)

PATIENT

- Name: Kevin
- Age: 46
- DOB: 1970-11-20
- Gender: Male
- Main Diagnosis:-
- Mobility: walking-without-aid
- Special Note: -

SERVICE PROVIDER

Provider is not assigned

[+](#) [-](#)

CLIENT

- Name: Kevin 1
- Mobile: +6588695610
- Email: kevinclient@gmail.com